Idea Proposal

MediFind AI

AI-Powered Medicine Availability & Alternative Finder

Team Details

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1. Problem Statement

Patients frequently need to visit multiple pharmacies to locate a prescribed medicine because there is currently no simple, reliable way to check medicine availability in advance. This results in wasted time, unnecessary travel costs, and delays in treatment — a problem that becomes especially critical during medical emergencies or for patients managing chronic conditions who depend on timely access to their medication.

2. Proposed Solution

MediFind AI is a machine learning-based web application that predicts the likelihood of a medicine being available at nearby pharmacies. Users can search for a medicine, view pharmacies ranked by their predicted probability of having it in stock, compare prices across pharmacies, and view verified generic alternatives. The platform also forecasts future medicine demand, helping pharmacies plan inventory more effectively and reduce stock-outs.

3. Objectives

- Predict the availability of a given medicine at nearby pharmacies using machine learning.
- Help users quickly identify pharmacies with a high probability of having the required medicine in stock.
- Enable price comparison for a medicine across multiple pharmacies.
- Display verified generic alternatives for informational purposes.
- Forecast future medicine demand to assist pharmacies with inventory planning.

4. Machine Learning Components

Availability Prediction: A classification model (Random Forest / XGBoost) that predicts the probability of a pharmacy having a given medicine in stock, based on features such as pharmacy size, medicine category, past restocking patterns, and demand trends.

Demand Forecasting: A time-series forecasting model (e.g., Prophet or regression-based) that estimates future demand for a medicine, supporting proactive inventory planning for pharmacies.

Medicine Name Correction: Fuzzy string matching (e.g., using RapidFuzz) to correctly identify medicines even when users misspell or partially type the name.

Prescription OCR (Future Scope): Optical character recognition to extract medicine names directly from uploaded prescription images, deferred from the initial build due to the complexity of handwriting recognition.

5. Proposed Features

- Medicine search with typo-tolerant (fuzzy) matching
- Nearby pharmacy locator using map integration
- AI-based availability prediction, ranking pharmacies by likelihood of stock
- Price comparison across pharmacies
- Generic medicine information for informational reference
- Demand forecasting dashboard for pharmacy-side inventory planning

6. Technology Stack

Frontend	React.js, Tailwind CSS
Backend	FastAPI (Python)
Database	PostgreSQL / Supabase
Machine Learning	Python, Scikit-learn, Pandas, XGBoost, Prophet
Maps & Location	Google Maps API / OpenStreetMap (Leaflet.js)
Deployment	Vercel (Frontend), Render (Backend), Supabase (DB)

7. Expected Outcome

The system will reduce the time and effort required for patients to locate medicines, improve overall user convenience, and help pharmacies optimize stock management through AI-driven demand prediction. While the initial prototype will rely on carefully designed synthetic inventory data — since real-time pharmacy inventory data is not publicly available — the system architecture is designed to integrate with real pharmacy inventory systems in the future, with all data assumptions clearly documented.

8. Future Scope

- Real-time integration with live pharmacy inventory systems
- Online medicine reservation and order-ahead functionality
- Integration with hospitals and clinics for prescription-linked search
- Multilingual support for wider accessibility
- AI chatbot for medicine-related queries
- Mobile application for Android and iOS